



C.4 Standing waves and resonance

Wave behaviour · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A standing (stationary) wave forms when two travelling waves of equal frequency and amplitude move in opposite directions and superpose, creating fixed points of no displacement (nodes) and points of maximum displacement (antinodes) that do not move along the medium, unlike a travelling wave. On a string fixed at both ends, resonant standing wave patterns occur at frequencies $f_n = nv/(2L)$ ($n = 1, 2, 3, \dots$), and in a pipe closed at one end, only odd harmonics occur, $f_n = nv/(4L)$ ($n = 1, 3, 5, \dots$), while an open pipe supports all harmonics $f_n = nv/(2L)$. Resonance occurs when a system is driven at (or near) one of its natural frequencies, producing a large amplitude response; damping reduces this amplitude and broadens the resonance peak.

Worked example: follow the reasoning

A string of length 0.80 m fixed at both ends vibrates in its second harmonic ($n=2$) with wave speed $60 \text{ m}\cdot\text{s}^{-1}$ on the string. Find the frequency.

$$f_n = nv/(2L) = 2(60)/(2 \times 0.80) = 120/1.6 = 75 \text{ Hz.}$$

Answer: $f = 75 \text{ Hz}$.

Quick check · multiple choice

A pipe closed at one end resonates at its fundamental frequency when its length is 0.50 m and the speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$. What is this fundamental frequency?

- A 85 Hz
- B 170 Hz
- C 340 Hz
- D 680 Hz

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A string fixed at both ends has length 1.2 m and is under tension so that the speed of transverse waves on it is $40 \text{ m}\cdot\text{s}^{-1}$. [7]
 - (a) Calculate the fundamental (first harmonic) frequency of vibration.
 - (b) Calculate the frequency of the third harmonic.
 - (c) Sketch (describe in words) the standing wave pattern for the third harmonic, stating the number of nodes and antinodes (excluding/including the fixed ends as appropriate).

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $f_1 = v/(2L) = 40/(2 \times 1.2) = 40/2.4 = 16.7 \text{ Hz}$.

(b) $f_3 = 3f_1 = 3 \times 16.7 = 50.0 \text{ Hz}$.

(c) The third harmonic has 3 antinodes and 4 nodes total, including the two fixed ends as nodes (the pattern shows one and a half full wavelengths fitted along the string, i.e. 3 loops between the two fixed-end nodes).

Quick check answer

B. For a closed pipe, $f_1 = v/(4L) = 340/(4 \times 0.50) = 340/2.0 = 170 \text{ Hz}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.